

Basic Counting

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Choosing two representatives

A club has 40 members. How many ways can it choose two representatives?

Choosing a first member and then a second gives

$$40 \cdot 39 = 1560.$$

But Alice followed by Bob and Bob followed by Alice give the same pair. Each pair appears twice, so the answer is

$$\frac{40 \cdot 39}{2} = 780.$$

Whether order matters changes what we count.

Finite sets

A **set** is a collection of distinct elements. For a finite set A , its **cardinality** $|A|$ is its number of elements.

$$A = \{a, b, c\}, \quad |A| = 3.$$

We write $S \subseteq A$ when every element of S belongs to A . We call S a **subset** of A .

$$\{a, c\} \subseteq A, \quad \{a, c\} = \{c, a\}.$$

The **empty set** $\emptyset$ contains no elements. Both $\emptyset$ and A itself are subsets of A .

Product rule

Suppose an outcome consists of two successive choices. There are m possibilities for the first choice and n for the second, for every first choice.

The **product rule** gives mn outcomes. For example, 3 shirts and 2 pairs of trousers give

$$3 \cdot 2 = 6 \text{ outfits.}$$

For k successive choices with counts $n_1, \dots, n_k$, the count is

$$n_1 n_2 \cdots n_k.$$

At each step, the stated count must hold for every earlier choice.

Strings with repetition

A **string of length** k is an ordered sequence of k symbols. Suppose each position can use any of n symbols, with repetition allowed.

The product rule gives

$$\underbrace{n \cdot n \cdots n}_{k \text{ positions}} = n^k.$$

A four-digit code uses symbols $0, 1, \dots, 9$ and allows leading zeros. There are $10^4 = 10,000$ codes, including 0000 and 0012.

Sum rule

Two classes are **disjoint** if they share no outcomes. If disjoint classes contain m and n outcomes, the **sum rule** gives $m + n$ outcomes in total.

An ID consists of either two or three letters from the 26 English letters. Repetition is allowed.

$$26^2 + 26^3 = 18,252.$$

The classes are disjoint because an ID cannot have both lengths. Multiplication counts successive choices. Addition combines disjoint cases.

Permutations

A **permutation** is an ordering of all the objects in a collection of distinct objects.

$$abc, acb, bac, bca, cab, cba$$

are the permutations of a, b, c .

For n distinct objects, the successive positions have $n, n - 1, \dots, 1$ choices. Define the **factorial** by

$$n! = n(n - 1) \cdots 2 \cdot 1 \quad (n \geq 1), \quad 0! = 1.$$

Therefore there are $n!$ permutations. In particular, $3! = 6$.

Ordered selections

Select and order k of n distinct objects, without repetition, where $0 \leq k \leq n$ are integers.

For $k \geq 1$, the product rule gives

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

Ten runners compete for first, second, and third place, with no ties. The number of possible podium orders is

$$10 \cdot 9 \cdot 8 = 720.$$

For $k = 0$, there is one empty selection, consistent with the formula.

Does repetition matter?

Form strings of length 3 using A, B, C, D, E . Order matters in both cases.

Repetition allowed: AAA and ABA are possible.

$$5^3 = 125.$$

No repetition: every position uses a different symbol.

$$5 \cdot 4 \cdot 3 = 60.$$

The second count decreases the available choices after each selection.

Combinations

A **combination** chooses objects without recording their order.
Choosing k of n distinct objects means choosing a k -element subset.
Write the number of these subsets as

$$\binom{n}{k},$$

read “ n choose k .” This is a **binomial coefficient**.
For example, choosing three medalists counts the set $\{a, b, c\}$ once,
regardless of who finishes first, second, or third.

Why divide by $k!$?

Each set of k distinct objects has exactly $k!$ orders. Thus we can count ordered selections by first choosing the set, then choosing its order:

$$\binom{n}{k} k! = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

Hence

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

For the ten runners, the 720 podium orders give $\binom{10}{3} = 720/3! = 120$ sets of medalists.

Boundary values and symmetry

For every integer $n \geq 0$,

$$\binom{n}{0} = \binom{n}{n} = 1.$$

There is one way to choose no elements and one way to choose all elements.

Choosing k elements also determines the $n - k$ elements left out, so

$$\binom{n}{k} = \binom{n}{n-k} \quad (0 \leq k \leq n).$$

For instance, choosing 8 of 10 people is equivalent to choosing the 2 people left out: $\binom{10}{8} = \binom{10}{2} = 45$.

Choosing a team

A class has 5 first-year students and 7 second-year students. How many teams contain exactly 2 first-year and 3 second-year students?

Choose each part of the team:

$$\binom{5}{2} = 10, \quad \binom{7}{3} = 35.$$

Each first-year pair can accompany any second-year triple. The product rule gives

$$\binom{5}{2} \binom{7}{3} = 350 \text{ teams.}$$

There are no roles or rankings within either group.

Number of subsets

Let $A = \{a_1, \dots, a_n\}$ have n distinct elements. To choose a subset, include or exclude each element.

Each of the n choices has two possibilities, giving

2^n subsets.

A **bit** is 0 or 1. With a fixed element order, a bit string records exclusion by 0 and inclusion by 1. For $A = \{a, b, c\}$,

$$\{a, c\} \longleftrightarrow 101, \quad \emptyset \longleftrightarrow 000.$$

The count includes the empty subset and the whole set.

Counting subsets by size

We can count all subsets of an n -element set by their sizes. There are $\binom{n}{k}$ subsets of size k .

Every subset has exactly one size, so the sum rule gives

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Equivalently,

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Counting the same collection in two ways is called **double counting**.

Pascal's identity

Fix one element a of an n -element set, where $1 \leq k < n$. A k -element subset either contains a or leaves it out.

- ▶ Contains a : choose the other $k - 1$ elements from the remaining $n - 1$.
- ▶ Leaves out a : choose all k elements from the remaining $n - 1$.

Adding the two disjoint cases gives

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

For example, $\binom{5}{2} = \binom{4}{1} + \binom{4}{2} = 4 + 6 = 10$.

Choosing from the factors

For real numbers x, y , consider

$$(x + y)^4 = (x + y)(x + y)(x + y)(x + y).$$

Expanding means choosing either x or y from each factor.

To obtain x^2y^2 , choose the two factors that contribute y . There are $\binom{4}{2} = 6$ choices, so the coefficient is 6.

More generally, among n factors, exactly $\binom{n}{k}$ choices produce $x^{n-k}y^k$.

Binomial theorem

For real numbers x, y and an integer $n \geq 1$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

The integer k counts how many factors contribute y . Choosing those factors explains each coefficient.

For $n = 3$,

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3.$$

Finding a coefficient

What is the coefficient of x^2y^3 in $(2x + y)^5$?

Choose y from 3 of the 5 factors and $2x$ from the other 2:

$$\binom{5}{3} (2x)^2 y^3 = 10 \cdot 4x^2 y^3 = 40x^2 y^3.$$

Thus the coefficient is 40.

As another application, setting $x = y = 1$ gives

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k},$$

agreeing with our count of subsets.

Order and repetition

Use the five symbols A, B, C, D, E to make a selection of size 3.

Outcome	Example	Count
Ordered, repetition allowed	ABA	$5^3 = 125$
Ordered, no repetition	ABC	$5 \cdot 4 \cdot 3 = 60$
Unordered, no repetition	$\{A, B, C\}$	$\binom{5}{3} = 10$

In the last row, changing the order does not create another outcome.

Summary

For $n \geq 1$ distinct objects and an integer $0 \leq k \leq n$:

What we count	Number
Ordered choices, repetition allowed	n^k
Ordered choices, no repetition	$n!/(n-k)!$
Unordered choices, no repetition	$\binom{n}{k}$
All subsets	2^n

The product rule combines successive choices. The sum rule combines disjoint cases. Decide whether order and repetition matter before counting.